

MA1521 Calculus for Computing

Lecture Group 2

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Lesson 10: Friday 11 September 2026

(Week 5)

Remark. Q09.8 to Q09.10 are leftover exercises from the Tuesday class.

Remark. Q10.1 to Q10.9 are on on Sections 13.4, 13.5, 13.6 and 13.9.

Q09.8. Let $f(x, y, z) = xy^2z^3$ and $P = (1, -2, 1)$. Find the rate of change of f at P in the direction $\mathbf{u} = \left\langle \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$.

[Remark: The question is the same as asking to find the directional derivative $D_{\mathbf{u}}f(P)$).

Q09.9. Let a be a positive number. Consider the function

$$f(x, y) = (x^2 + a) e^{2x+3y^2}.$$

It is given that the directional derivative of f at the point $(-6, 2)$ in the direction of the vector $\mathbf{i} + 2\mathbf{j}$ is $216\sqrt{5}$. Find the value of a .

Q09.10. Consider the function $f(x, y) = 2ye^{-x} + 1$ and the point $P(0, -2)$.

(a) Find the directional derivative of f at P in the direction of each of the following vectors:

(i) $\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$, (ii) $12\mathbf{i} + 5\mathbf{j}$.

(b) Find the direction that gives the *largest possible* directional derivative of f at P .

(c) Find the direction along which f decreases most rapidly at P .

(d) Find the directions along which f exhibits no change at P .

Q10.1. Consider the surface in the xyz -space defined by the equation

$$\cos(\pi x) - x^2 y + e^{xz} + yz = 4.$$

Find equations for the tangent plane and normal line to the surface at the point $(0, 1, 2)$.

Q10.2. Let $\mathbf{C}$ be the curve of the intersection of the two surfaces $x^3 + 2xy + yz = 7$ and $3x^2 - yz = 1$.

Find parametric equations of the tangent line to $\mathbf{C}$ at $(1, 2, 1)$.

[Suggestion: First, for each of the two surfaces, find a normal vector of the surface at the point $(1, 2, 1)$.

Q10.3. Let $\mathbf{S}$ be the surface given by the equation $z = \mathbf{a}x^2 + \mathbf{b}y^2$, where $\mathbf{a}$ and $\mathbf{b}$ are some positive constants.

Suppose normal lines are drawn at all the points on $\mathbf{S}$ that are at height $\mathbf{h}$ above the $\mathbf{xy}$ -plane.

Find an equation of the curve $\mathbf{C}$ (in terms of $\mathbf{x}$ and $\mathbf{y}$) consisting of all the intersection points of these normal lines and the $\mathbf{xy}$ -plane.

Q10.4. Consider the function $\mathbf{f}(\mathbf{x}, \mathbf{y}) = \mathbf{e}^{2\mathbf{x}+\mathbf{y}}$.

(a) Find the linearization of $\mathbf{f}$ at the point $(\mathbf{0}, \mathbf{0})$.

(b) Find the linearization of $\mathbf{f}$ at the point $(\mathbf{1}, \mathbf{2})$.

[Remark: In both (a) and (b), give your answers in the form $\mathbf{a}\mathbf{x} + \mathbf{b}\mathbf{y} + \mathbf{c}$.]

Q10.5. Consider the function $w = xy^2z^3$. Find the (total) differential dw in terms of dx , dy and dz .

Q10.6. Consider the function $w = w(x, y, z)$ defined implicitly by the equation

$$z^2 + xyw + yw^3 = 0.$$

Find the differential dw in terms of dx , dy and dz .

[Suggestion: Regard both sides as functions of x, y, z, w , and take differentials on both sides. Then solve for dw .]

[Remark. Alternatively, you may also use implicit differentiation to find w_x, w_y, w_z and then get dw .]

Q10.7. Consider the function $f(x, y, z) = xz + \sin(xyz)$.

Using linear approximation, estimate the change in the value of f when (x, y, z) changes from $(\frac{1}{2}, \pi, \frac{1}{3})$ to

$(\frac{1}{2} + 0.03, \pi + 0.04, \frac{1}{3} - 0.02)$.

Q10.8. Let $f(x, y)$ be a function, and (x_o, y_o) be a point in the domain of f . Consider the quadratic polynomial (in two variables) given by

$$\begin{aligned} g(x, y) = & a + b(x - x_o) + c(y - y_o) \\ & + d(x - x_o)^2 + e(x - x_o)(y - y_o) \\ & + f(y - y_o)^2, \end{aligned}$$

where a, b, c, d, e, f are constants.

Determine a, b, c, d, e, f (in terms of data on f) if

$$\begin{aligned} g(x_o, y_o) &= f(x_o, y_o), & g_x(x_o, y_o) &= f_x(x_o, y_o) \\ g_y(x_o, y_o) &= f_y(x_o, y_o), & g_{xx}(x_o, y_o) &= f_{xx}(x_o, y_o), \\ g_{xy}(x_o, y_o) &= f_{xy}(x_o, y_o), & \text{and } g_{yy}(x_o, y_o) &= f_{yy}(x_o, y_o). \end{aligned}$$

Hence write down $g(x, y)$.

[Remark: $\mathbf{g}(\mathbf{x}, \mathbf{y})$ give a quadratic approximation of $\mathbf{f}(\mathbf{x}, \mathbf{y})$ for $(\mathbf{x}, \mathbf{y})$ near $(\mathbf{x}_o, \mathbf{y}_o)$. Also, $\mathbf{g}(\mathbf{x}, \mathbf{y})$ is called the 2^{nd} -degree Taylor polynomial of $\mathbf{f}$ at $(\mathbf{x}_o, \mathbf{y}_o)$.]

Q10.9. Consider the function $\mathbf{f}(\mathbf{x}, \mathbf{y}) = \mathbf{e}^{\mathbf{x}+4\mathbf{y}} \cos \mathbf{x}$.

(a) Find the second degree Taylor polynomial of $\mathbf{f}$ at $(\mathbf{0}, \mathbf{0})$.

(b) Using part (a), find an approximate value of $\mathbf{f}(\mathbf{0.2}, \mathbf{0.1})$.